ECON 401: Advanced Econometrics

Panel Data Models, Quasi-Experiments, and Causal Inference

Department of Economics • Fall Semester 2026

Instructor: Prof. Elena Rostova
Email: `rostova@university.edu`
Office Hours: Wednesday 10:00–12:00
Lecture Hours: Tue/Thu 13:30–15:00
Room: Economics Hall 310
Lab Session: Thursday 16:00–18:00 (Lab B)

Course Overview & Objectives

ECON 401 equips students with modern econometric tools for causal identification using observational and quasi-experimental data. Key topics include potential outcomes, Two-Way Fixed Effects, Difference-in-Differences, Instrumental Variables (2SLS), and cluster-robust inference.

Grading Scheme

Assessment Component	Weight	Description
Problem Sets (5)	30%	Econometric proofs and empirical Python/Stata replications
Midterm Examination	25%	Theoretical derivations and identification analysis
Empirical Term Paper	35%	Original causal estimation project using real-world microdata
Class Participation	10%	Discussion of landmark empirical papers

15-Week Curriculum Roadmap

- **Weeks 1–3:** Potential outcomes framework, SUTVA, and selection on observables.
- **Weeks 4–6:** Panel data models: Pooled OLS, Random Effects, and Within-Group Fixed Effects.
- **Weeks 7–8:** Difference-in-Differences, parallel trends, and staggered rollouts.
- **Week 9: Midterm Examination**.
- **Weeks 10–12:** Instrumental variables, 2SLS, weak instrument diagnostics, and LATE.
- **Weeks 13–15:** Cluster-robust inference, Liang-Zeger standard errors, and bootstrap inference.